becomes much easier when you have a queue that can buffer requests for a while. Therefore, you will learn how to transition to an asynchronous process with the help of SQS next.

14.2.1 Turning a synchronous process into an asynchronous one

A typical synchronous process looks like this: a user makes a request to your web server, something happens on the web server, and a result is returned to the user. To make things more concrete, we'll talk about the process of creating a preview image of a URL in the following example, illustrated in figure 14.5:

1. The user submits a URL.
2. The web server downloads the content at the URL, takes a screenshot, and renders it as a PNG image.
3. The web server returns the PNG to the user.



Figure 14.5 A synchronous process to create a screenshot of a website.

With one small trick, this process can be made asynchronous and benefit from the elasticity of a message queue, for example, during peak traffic, as shown in figure 14.6:

1. The user submits a URL.